1

Compatctness

1.1 Basic Definition and Properties

Definition 1.1.1

A collection $\mathcal{U}$ of open subsets of a space X is called an ***open cover*** for X if and only if the union of sets in $\mathcal{U}$ contains X. The space X is ***compact*** if and only if every open cover of X has a finite subcover.

Theorem 1.1.2

If X is compact and $f : X \to Y$ is continuous, then fX is compact.

Theorem 1.1.3 ▸ Bolzano-Weierstrass

Every infinite set in a compact space has a limit point.

1.2 FIP , a dual description

There is a dual description.

Definition 1.2.1

Let S be a collection of sets. We say that S has the ***finite intersection property*** if and only if for every finite subcollection $A_1, \dots, A_n \subseteq S$, the intersection $A_1 \cap \cdots \cap A_n \neq \varnothing$. We abbreviate the finite intersection property by FIP.

Theorem 1.2.2

A space X is compact if and only if every collection of closed subsets of X with the FIP has nonempty intersection.

Proof. Let $\{F_\alpha\}_{\alpha \in A}$ be any collection of closed subsets of X with the FIP. If $\bigcap_\alpha F_\alpha = \varnothing$, then $\{F_\alpha^c\}_{\alpha \in A}$ is an open cover of X, which has a subcover $F_1^c, F_2^c, \cdots, F_m^c$, such that

$$\bigcap_{k=1}^{m} F_m = \left(\bigcup_{k=1}^{m} F_k^c\right)^c = X^c = \varnothing$$

a contradiction.

Conversely, if every closed collection is with FIP. Then, for each open cover $\{U_\alpha\}_{\alpha\in A}$ of X, if $\{U_\alpha\}_{\alpha\in A}$ dose not have a finite subcover of F_α, $\{U_\alpha^c\}_{\alpha\in A}$ is a closed collection of X with the FIP, which imlies that $\bigcap U_\alpha^c \neq \varnothing$, which is a contradiction to $\bigcup U_\alpha = X$. □

Theorem 1.2.3

Closed subsets of compact spaces are compact.

1.3 Compactness and Hausdorff

Theorem 1.3.1 ▶ Seperate a point and a compact set

Let X be Hausdorff. For any point $x \in X$ and any compact set $K \subseteq X\backslash\{x\}$ there exist disjoint open sets U and V with $x \in U$ and $K \subseteq V$.

Proof. □

Corollary 1.3.2 ▶ Compact subsets of Hausdorff spaces are closed

Compact subsets of Hausdorff spaces are closed.

Corollary 1.3.3 ▶ A map from compact to Hausdorff spaces is closed

If X is compact and Y is Hausdorff, then every map $f : X \to Y$ is closed.[1]
In particular,

- if f is injective, then it is an embedding;
- if f is surjective, then it is a quotient map;
- if f is bijective, then it is a homeomorphism.

1.4 Local Compactness

Definition 1.4.1

A space X is said to be ***locally compact at x*** if there is some compact subspace C of X that contains a neighborhood of x. If X is locally compact at each of its points, X is said simply to be ***locally compact***.

Theorem 1.4.2

Let X be a space. Then X is ***locally compact Hausdorff*** if and only if there exists a space Y satisfying the following conditions:

1. X is a subspace of Y.
2. The set $Y - X$ consists of a single point.
3. Y is a compact Hausdorff space.

If Y and Y' are two spaces satisfying these conditions, then there is a homeomorphism of Y with Y' that equals the identity map on X.

Proof. □